

Machine Learning

Module 1:

Introduction:

1. What is machine learning? Explain briefly the different types of machine learning.
2. How is supervised learning different from unsupervised learning?
3. How is regression different from classification?
4. What do you mean by overfitting and underfitting of a model?
5. What is cost function in machine learning?
6. What is training set and test set in machine learning model?
7. What is the trade-off between an Overfitting and underfitting model in regression modeling?
8. What is the Euclidian distance between two data points: A (1,3) and B(2,3)?
9. What is training set and test set in machine learning model?
10. Write a short note on trade-off between bias and variance in machine learning.

Supervised Learning:

Linear Regression:

1. What is a linear regression algorithm? How do you interpret a linear regression model?
2. Numerical question on simple linear regression - practice

Logistic Regression:

1. What is Logistic Regression?

2. What is Logistic Function or Sigmoid Function? Write its role in logistic regression.
3. Why does logistic regression come under the category of classification problems?
4. How is logistic regression different from linear regression?

KNN:

1. Which machine learning algorithm is called lazy learner and why?
2. Write the advantages and disadvantages of KNN algorithm.
3. Why is KNN a non-parametric Algorithm?
4. Why is the odd value of “K” preferred over even values in the KNN algorithm?
5. How does the KNN algorithm make the predictions on the unseen dataset?
6. Write down the steps of KNN algorithm.
7. Numerical question on KNN.

Naïve Bayes Classifier:

1. Define Bayes theorem. What is the basic assumption in Naïve Bayes classifier?
2. Discuss the advantages and limitations of Naive Bayes algorithm.
3. Numerical question

Decision Tree:

1. Explain the terms Information Gain and Gini index with the help of an example.
2. Construct a decision tree Training data set is given.

3. What is a decision tree? Explain with an example.
4. Write down the ID3 algorithm.
5. Advantages and disadvantages of Decision Tree.
6. Numerical question

SVM:

1. Explain the working principle of support vector machine algorithm.
2. Write the difference between linear and non-linear SVM.
3. What is the Euclidian distance between two data points?
4. Numerical question (Linear SVM)

Confusion Matrix:

1. Confusion matrix will be given. Find out the accuracy, misclassification rate, sensitivity and other metrics.
- 2.

		Predicted Class			
		System A		System B	
Actual Class		Negative	Positive	Negative	Positive
	Negative	80	20	85	15
	Positive	25	75	30	70

Find out the number of False Positives (FP) and False Negatives (FN) for both systems.

Module 2:

Unsupervised Learning:

1. Write the differences between supervised and unsupervised learning.
2. Write down the differences between classification and clustering.

3. What do you mean by clustering? Write note on the different types of clustering.

4. What are the different types of unsupervised learning?

k-means algorithm:

1. Explain briefly the working principles of k-means algorithm.
2. What are the differences between K-means clustering and K-nearest neighbor (KNN).
3. Explain the strengths and weaknesses of the k-means algorithm.
4. Differences between K-means clustering and K-medoids.
5. Numerical on K-means algorithm (practice 1D as well as 2D data points)

PCA:

1. Explain the term "Curse of Dimensionality". What are the ways of dealing with sparse features?
2. What are the common problems with sparse dataset in machine learning?
3. Why is machine learning difficult with sparse features?
4. What is difference between sparse data and missing data?
6. What is principal component analysis?
7. What are the steps involved in PCA algorithm.
8. What are the advantages and disadvantages of PCA algorithm?

Matrix Factorization:

1. Write a short note on matrix factorization.
2. Where Matrix Factorization is used?

Module 3:

1. What do you mean by Ensemble learning? Explain.
2. What are the advantages and disadvantages of ensemble learning?
3. How can we classify the various ensembles learning technique?
4. Compare bagging and boosting.
5. What is the principle behind the Random Forest ensemble model?
6. What are the advantages and disadvantages of the Random Forest algorithm?
7. Explain the differences between Random Forest and Decision Tree?
8. What is the general idea of boosting technique?
9. Compare AdaBoost and Gradient Boosting.
10. Explain the different advanced ensemble techniques.

Module 4:

1. Define Time series and time series data.

2. What are the different methods for time series forecasting?
3. Explain with examples the different component of time series.
4. Explain simple moving average technique with an example.
5. Write a brief note on seasonal variations.
6. Discuss the working strategy of semi-supervised learning.
- 7.State the difference between Semi-supervised and Reinforcement learning.
- 8.Discuss advantages and disadvantages of Reinforcement Learning.
- 9.Explain the architecture of an artificial neural network (ANN).
- 10.What is the role of the Activation functions in Neural Networks?
- 11.Explain the various types of activation functions available.
- 12.What are Feed Forward Networks? Explain.
13. Define deep learning. Name few deep learning algorithms.
How deep learning differs from machine learning?

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